

4(a)	<i>any one from:</i> <i>Mycobacterium tuberculosis ;</i> <i>Mycobacterium bovis ;</i>	1
4(b)	length X–Y (in mm) $\times 1000 \div 21\,000$; 1.7–1.8 (μm) (to 2 significant figures) ;	2
4(c)	<i>any two from:</i> peptidoglycan cell wall ; circular DNA ; naked DNA / DNA without histones ; 70S ribosomes ; DNA not enclosed by nuclear envelope ; A no (true) nucleus absence of, (double) membrane-bound organelles / named example of (double) membrane-bound organelle ; AVP ;	2
4(d)	<i>any three from:</i> <i>does not support:</i> 1 much larger proportion do not have HIV / AIDS / AW ; 2 data to support ; e.g. deaths of people from TB with HIV / AIDS is 0.25 million out of 1.5 million 3 other factors may cause more deaths from TB ; <i>supports:</i> 4 large proportion of deaths of people with HIV / AIDS are caused by TB ; 5 data to support ; e.g. deaths of people with HIV/AIDS caused by TB is a minority / 0.25 million out of 0.75 million 6 fewer people will be immune compromised ; 7 AVP ; e.g. ref. to different modes of transmission so, cases of / deaths from, TB may not change not enough information, qualified	3

Question	Answer	Marks
4(e)	any three from: 1 ref. to opportunistic infections causing death ; 2 low levels of / less, cytokine, secreted / AW ; 3 two from - activity of macrophages not stimulated - B-lymphocyte / plasma cells / humoral response, not stimulated A activate B-lymphocytes / activate B cells (so) lower concentrations of / less / no, antibody, produced / secreted ; - stimulates, T-cytotoxic / T-killer, cells ; 4 (so) fewer, bacteria killed / viruses eliminated / pathogens killed ; 5 fewer memory cells (to fight future infection) ; 6 AVP ; e.g. idea of overburdened immune system	3

Question	Answer	Marks
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